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# Time-lag correction for oceanic in situ CO<sub>2</sub> sensors using a physics-informed hybrid BiLSTM-attention and Kalman filter framework

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## Abstract

High spatiotemporal-resolution observations of the oceanic partial pressure of carbon dioxide ( $p\text{CO}_2$ ) are essential for advancing our understanding of the global carbon cycle. Nonetheless, oceanic in situ  $p\text{CO}_2$  sensors mounted on mobile platforms frequently encounter challenges such as a physical response lag when crossing concentration gradients. To address this issue and consider the inherent noise of the sensors as well as data heterogeneity, including inconsistent data lengths, we propose a physics-informed deep learning framework for sensor time-lag correction. A single-layer bidirectional long short-term memory (BiLSTM) network coupled with an attention mechanism was designed to capture transient concentration gradients without oversmoothing the signal. A Kalman filter was then applied as a post-processing module to suppress the prediction noise without inducing a secondary physical lag. The experimental results demonstrate that this framework corrects the phase delay while minimizing the residual errors. Quantitative results show that the proposed method outperforms the physical linear time-invariant (LTI) model, linear autoregressive with exogenous input (ARX) model, and traditional statistical baselines (e.g., support vector regression, SVR). It dynamically corrects phase delay and reduces system response time by more than 80% without amplifying high-frequency noise, thereby offering a computationally efficient solution for continuous ocean observation.

**Keywords** Marine CO<sub>2</sub> sensor, Time-lag correction, Physics-informed deep learning, Kalman filter, BiLSTM-attention

## 1 Introduction

Oceans, the largest global carbon reservoirs, play a critical role in moderating climate change by absorbing a large fraction of anthropogenic carbon dioxide (CO<sub>2</sub>) emissions. However, this uptake drives ocean acidification and poses serious risks to marine ecosystems (Doney

et al. 2009; Gruber et al. 2023). Accurate quantification of long-term trends and variability in ocean carbon sinks requires high-resolution sustained observations (Waninkhof et al. 2013; Gruber et al. 2023).

Most in situ marine  $p\text{CO}_2$  sensors in widespread use operate via gas-liquid equilibration across a semi-permeable membrane. The inherent physical permeation process underlying these sensors induces a measurable response lag before the instrument reaches equilibrium (Fietzek et al. 2014). In static or low-flow environments, this lag is often negligible; however, it becomes problematic for mobile platforms that sample rapidly changing water masses. Modern observational programs increasingly rely on autonomous, mobile platforms such as biogeochemical-Argo (BGC-Argo) floats, autonomous underwater vehicles (AUVs), underwater gliders, and

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similar systems to achieve three-dimensional, basin-scale coverage. Large initiatives such as the Southern Ocean Carbon and Climate Observations and Modeling (SOC-COM) project depend heavily on profiling floats to capture year-round biogeochemical dynamics across wide regions (Sarmiento et al. 2023). When these high-speed mobile platforms traverse sharp spatiotemporal  $p\text{CO}_2$  gradients (e.g., ocean fronts or eddies), sensor hysteresis produces a pronounced temporal smearing effect that displaces measured values in space and time and can bias carbon flux estimates (Bittig et al. 2014; Mihaylov and El Naggar 2021). Consequently, untethered mobile observation platforms require sensors or correction methods that provide rapid dynamic responses without exceeding tight payloads, volumes, and energy budgets.

The observation platform type also strongly affects data quality. Fixed buoys and seabed observatories deliver high-frequency and long-duration records but are spatially constrained (Sutton et al. 2014). Mobile platforms extend spatial coverage, but impose stringent constraints on sensor size, weight, and power. Remotely operated vehicles (ROVs) used to survey seafloor vents and cold seeps can host relatively large, fast  $p\text{CO}_2$  systems that achieve second-level response times through auxiliary hardware. However, these solutions are often bulky and power-hungry (Zhang et al. 2017; Zhang et al. 2023). In contrast, Argo floats, gliders, and several AUVs must use miniaturized, low-power sensors that trade-off accuracy and dynamic response to meet payload (Roemmich et al. 2009; Rudnick 2016). As hardware miniaturization and response improvements encounter physical constraints, software-based compensation for the sensor dynamic lag has become an attractive and necessary alternative.

A complicating factor for software correction efforts is the heterogeneity of marine  $p\text{CO}_2$  time series. Differences among manufacturers, production batches, deployment ages, and usage histories produce datasets that exhibit heterogeneous time-series (HTS) characteristics, such as large variability in statistical properties (mean and variance) and uneven lengths, which challenge conventional processing and modeling approaches (Fawaz et al. 2019). Physics-based numerical methods such as linear time-invariant (LTI) inverse filtering have been used to reconstruct delayed signals. Although grounded in sensor physics, these methods tend to amplify high-frequency environmental noise when computing concentration gradients, thereby producing severe signal distortions.

Recent advances in data-driven and hybrid approaches have offered promising alternatives. Machine learning and intelligent optimization methods have shown superior performance over simple empirical corrections for complex and dynamic environmental signals (Rajwar

et al. 2023; Wang et al. 2024). In the emerging domain of the ‘Smart Ocean,’ artificial intelligence has been successfully applied to highly nonlinear oceanographic problems, including monitoring and adaptive exploration (Song et al. 2023). Sequence models such as long short-term memory (LSTM) networks augmented with attention mechanisms have been used to extract transient features and suppress fluid-dynamic disturbances for platforms such as underwater gliders (Tian et al. 2025). These successes motivated the use of physics-informed deep learning to resolve the trade-off between time-lag correction and noise amplification in mobile in situ  $p\text{CO}_2$  sensing.

In this study, we developed a physics-informed hybrid deep learning framework to overcome the limitations of traditional physical compensation. Our approach combines a single-layer bidirectional LSTM (BiLSTM) with an attention mechanism to capture transient concentration gradients without oversmoothing, and cascades a Kalman filter as a post-processing stage to suppress small-scale prediction jitter without introducing secondary physical lag. The main contributions of this study are as follows.

- 1) Hybrid correction framework: We propose a method integrating single-layer BiLSTM with attention that balances time-lag compensation and noise suppression, avoiding the high-frequency noise amplification typical of LTI inverse filtering, while correcting the phase delay and minimizing the residual error.
- 2) Demonstration of attention utility: Ablation experiments showed that the attention mechanism is essential for a shallow recurrent network to recognize rapid gradient changes, reduce delayed responses observed in standard recurrent architectures during sharp  $p\text{CO}_2$  fluctuations, and enable phase correction without excessive smoothing.
- 3) Kalman filter integration: Cascading a Kalman filter as a postprocessing module smoothens the high-frequency prediction variability from the neural network. Quantitatively, the combined method reduced the system response time by 72.8% for typical rising sequences and 43.5% for falling sequences while lowering the root mean square error (RMSE) by 12%–28%.

Combined, these results indicate that a physics-informed hybrid learning approach can substantially improve the dynamic fidelity of  $p\text{CO}_2$  measurements using miniaturized sensors on mobile oceanographic platforms, thereby enabling more accurate estimates of ocean carbon fluxes under realistic deployment constraints.

## 2 Related work

### 2.1 Time series analysis and the evolution of deep learning model

Time-series analysis remains a fundamental challenge in the scientific and engineering domains. Traditional methods, such as the autoregressive integrated moving average (ARIMA), often struggle to effectively capture the nonlinear and non-stationary characteristics typical of marine environments. Consequently, deep learning approaches have become increasingly prominent in this field (Lim and Zohren 2021). To address the vanishing gradient problem in conventional recurrent neural networks (RNNs), advanced architectures, including long short-term memory (LSTM) networks (Sherstinsky 2020) and gated recurrent units (GRUs), introduce gating mechanisms that enable the effective modeling of long-term dependencies (Fawaz et al. 2019; Hewage et al. 2021). Furthermore, bidirectional LSTM (BiLSTM) enhances contextual representation by incorporating information from both the past and future directions, thereby overcoming the limitations of unidirectional processing (Sagheer and Kotb 2019; Lin et al. 2023). The integration of attention mechanisms further improves the model performance by allowing a dynamic focus on critical historical information within sequences (Niu et al. 2021). Consequently, hybrid BiLSTM-attention architectures have demonstrated a strong capability to model complex and highly variable sequences (Xiang et al. 2020; Zhu et al. 2022).

Recent advances in time-series modeling and data augmentation techniques (e.g., Informer and PatchTST) have achieved state-of-the-art performance in long-sequence modeling (Xiang et al. 2020; Zhu et al. 2022). However, these models are typically heavily parameterized and require large-scale datasets for effective training to avoid overfitting. In marine sensor calibration scenarios, where data collection is costly and limited, acquiring large datasets is impractical. Therefore, the BiLSTM-attention framework offers a favorable balance between computational efficiency and predictive accuracy, particularly for small- to medium-sized datasets.

### 2.2 Methods for correcting sensor response lag

Sensor response lag presents a critical challenge for dynamic measurements. Due to physical constraints on hardware miniaturization, algorithmic ‘soft correction’ methods based on mathematical modeling have become the dominant approach. Classical correction techniques typically focus on the physical response mechanisms of sensors using differential equations or deconvolution algorithms for inverse compensation. For example, LTI system models are commonly employed to perform inverse filtering of aquatic oxygen and carbon dioxide

sensor data (Fiedler et al. 2013; Bittig et al. 2014; Gordon et al. 2020).

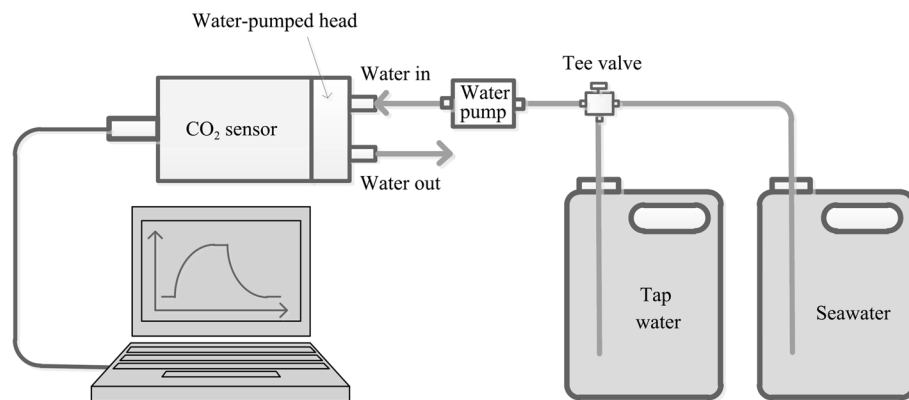
However, inverse filtering is highly sensitive to high frequency noise. The differentiation of fluctuating signals inherently amplifies noise, often resulting in distorted and oscillatory outputs. To mitigate these effects, alternative approaches such as autoregressive with exogenous input (ARX) models and Kalman filters have been explored (Tomczyk et al. 2022). Although these traditional methods can partially suppress noise, they often fail to achieve instantaneous phase correction simultaneously and rely heavily on accurately specified physical model parameters. Consequently, the fundamental trade-off between ‘time-lag elimination’ and ‘noise amplification’ remains unresolved. This limitation has motivated the adoption of physics-informed data-driven deep learning approaches.

### 2.3 Attention mechanism and post-processing micro-jitter suppression

In real-world marine environments, calibration data collected under varying experimental conditions exhibit significant heterogeneity in noise characteristics and distribution (Zhuang et al. 2021). Advanced approaches, such as transfer learning (Weber et al. 2021; Zhuang et al. 2021), domain adaptation (Wilson and Cook 2020), and meta-learning (Wang et al. 2020) have been widely proposed to address such heterogeneity. However, these methods typically require large datasets to achieve effective domain alignment. Given the high cost and strict control requirements for marine sensor calibration, acquiring such datasets is challenging.

Instead of relying on data-intensive approaches, this study directly addresses the core physical challenges of sensor lag and fluid-induced noise through a combination of attention mechanisms and Kalman filtering. When sensors mounted on mobile platforms traverse regions with sharp  $p\text{CO}_2$  gradients, standard recurrent models often fail to distinguish true concentration changes from background noise, leading to delayed correction. The attention mechanism serves as a dynamic feature selection module, enabling BiLSTM to assign greater importance to salient transition points while suppressing irrelevant or noisy historical information.

Although the attention-enhanced model effectively corrects the macroscopic phase lag, its nonlinear nature may introduce high-frequency micro-jitters into the predictions. To ensure the data quality required for marine carbon flux estimation, a Kalman filter was applied as a postprocessing step. As an optimal recursive smoother, it suppresses stochastic fluctuations without introducing additional lags. In addition, before training, each step-response sequence was temporally aligned to remove



**Fig. 1** Schematic of the response time testing setup for the underwater in situ  $\text{CO}_2$  sensor

initial fluid disturbances and ensure a stable baseline for the sliding-window learning process.

### 3 Methodology

#### 3.1 Experimental data acquisition and heterogeneity construction

The dataset employed in this study consisted of 13 independent marine  $\text{CO}_2$  time-series sequences collected using a low-cost in situ  $p\text{CO}_2$  sensor based on membrane inlet and non-dispersive infrared (NDIR) principles (Li et al. 2022), with a sampling frequency of 1 Hz. The experimental setup is shown in Fig. 1. A water-pumped head was positioned at front of the sensor (i.e., the permeable membrane side) and connected to a water pump and a tee valve through tubing. Tap water and seawater, which represent distinct  $p\text{CO}_2$  levels, were used as the input sources.

During the experiments, a tee valve was used to rapidly switch the inlet lines, artificially triggering sharp changes in the concentration step. This design simulates the sharp  $p\text{CO}_2$  concentration gradients frequently encountered by mobile observation platforms (e.g., underwater gliders and AUVs) as they traverse distinct ocean water masses. The recorded sequences included complete dynamic response cycles, that is, rising, falling, and steady-state phases, providing a rigorous and comprehensive basis for evaluating the lag correction capabilities and noise robustness of the models.

For supervised training, the target labels (i.e., ideal ground truth) must accurately reflect the true external physical processes. The ground truth is not an arbitrarily defined instantaneous square wave but a physically derived reference signal. Although water source switching at the valve is nearly instantaneous ( $< 1$  s), the actual  $p\text{CO}_2$  concentration inside the sensor's external water chamber changes continuously owing to fluid dynamics. The theoretical transition time is governed by the ratio

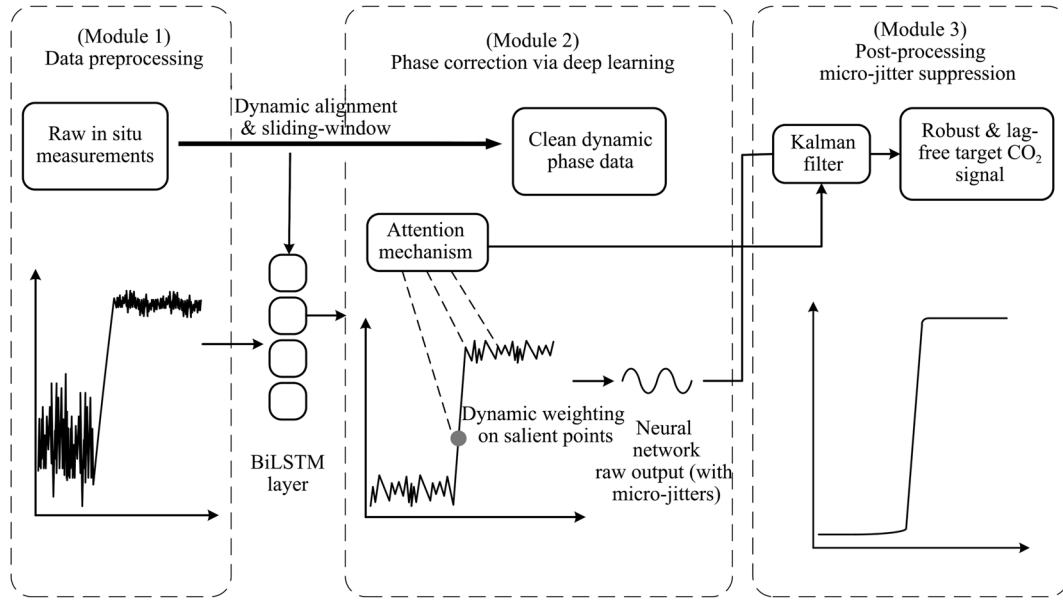
of dead volume to pump flow rate. Based on our experimental hardware parameters, the complete fluid replacement process required approximately 10 s. Accordingly, the ground truth was modeled as a continuous logistic step curve with approximately 10 s transition period. The model was trained to map raw sensor signals, whose equilibration exceeded 100 s owing to membrane permeation, to this physically realistic baseline.

In practical applications, marine in situ sensors require field calibration to ensure measurement accuracy. However, because of the limited time available for field deployment, the acquired calibration datasets were inherently constrained in size. This operational constraint necessitates a data-efficient modeling approach capable of robust performance with limited training samples. To authentically reflect real-world operational constraints, we constructed a compact laboratory dataset comprising 13 sequences.

#### 3.2 Data preprocessing and sliding-window segmentation

To address the sensor's physical response lag and suppress high-frequency fluid noise, we propose a hybrid BiLSTM-attention and Kalman filter framework. The overall architecture is shown in Fig. 2. Prior to training the deep learning model, a standardized data preprocessing pipeline was applied to ensure training stability and enhance feature extraction.

In practical deployments, marine in situ sensors require field calibration; however, the calibration time is often limited, resulting in small datasets. In this study, 13 independent step-response sequences were collected from the laboratory experiments. For supervised learning, 11 sequences were used for training, whereas two sequences were reserved as independent out-of-distribution test sets: one exhibiting a strictly decreasing trend (Set A) and the other a strictly increasing trend (Set B). These challenging cases were intentionally selected to



**Fig. 2** Flowchart of the proposed physics-informed hybrid correction framework, integrating dynamic feature extraction, BiLSTM-attention phase correction, and Kalman filter micro-jitter suppression. The grey point represents salient points

evaluate the generalization of the model under extreme transient conditions.

Owing to the variations in the sequence length and valve-switching onset, a temporal alignment strategy was applied to synchronize the transient events before inputting them into the model. Instead of a fixed-length truncation, the start of each step-response event was dynamically aligned. Continuous sliding-window sequence-to-sequence segmentation was then implemented to directly process the data stream, and the window length was set to 15 time steps ( $SEQ\_LEN = 15$ ). Given the sampling rate of 1 Hz and the fluid transition time of approximately 10 s within the chamber, this window size is physically well-matched; it captures the complete transient dynamics while avoiding redundancy from steady-state noise. This targeted preprocessing effectively bypasses the initial pump-induced fluid disturbances and allows the network to continuously extract localized historical features across the entire dynamic transition phase, rendering the framework completely agnostic of the total length of the raw sequences.

Furthermore, to train the supervised BiLSTM-attention network, the ideal ground truth (i.e., training labels) for each sequence was mathematically constructed. Based on the physical nature of the instantaneous valve-switching experiment, the ground truth is defined as a sharp step function connecting the initial and final steady-state concentrations, representing the true instantaneous transient dynamics of the external fluid.

### 3.3 BiLSTM-attention model architecture

BiLSTM processes the input sequence in both the forward and backward LSTM layers to capture bidirectional temporal dependencies. The calculation formulas for the LSTM unit at time step  $t$  are as follows:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f), \quad (1)$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i), \quad (2)$$

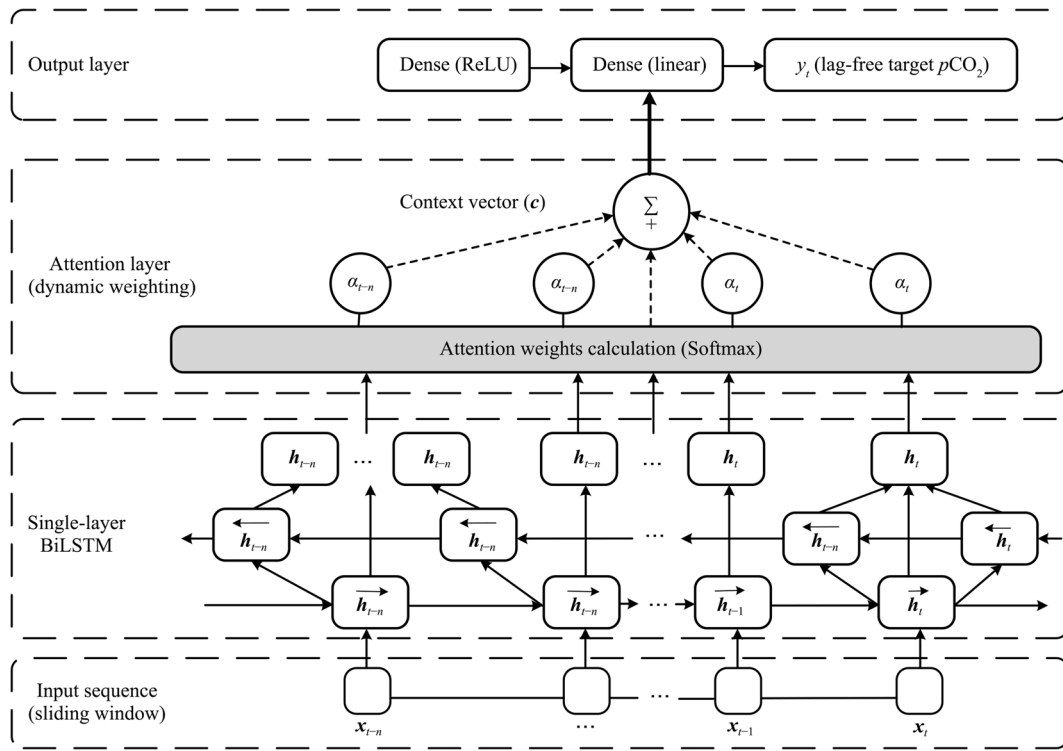
$$\tilde{\mathbf{C}}_t = \tanh(\mathbf{W}_C \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_C), \quad (3)$$

$$\mathbf{C}_t = \mathbf{f}_t * \mathbf{C}_{t-1} + \mathbf{i}_t * \tilde{\mathbf{C}}_t, \quad (4)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o), \quad (5)$$

$$\mathbf{h}_t = \mathbf{o}_t * \tanh(\mathbf{C}_t), \quad (6)$$

where  $\mathbf{x}_t$  denotes the input vector at time step  $t$ , and  $\mathbf{h}_{t-1}$  represents the hidden state vector from the previous time step. The terms  $\mathbf{f}_t$ ,  $\mathbf{i}_t$ , and  $\mathbf{o}_t$  refer to the activation vectors for the forget, input, and output gates, respectively.  $\tilde{\mathbf{C}}_t$  and  $\mathbf{C}_t$  represent the candidate and updated cell states, respectively.  $\mathbf{W}_f$ ,  $\mathbf{W}_i$ ,  $\mathbf{W}_C$ , and  $\mathbf{W}_o$  are the weight matrices, and  $\mathbf{b}_f$ ,  $\mathbf{b}_i$ ,  $\mathbf{b}_C$ , and  $\mathbf{b}_o$  denote the corresponding bias vectors. The symbol  $\sigma$  represents the sigmoid activation function, and  $*$  denotes the element wise multiplication. The final output of the BiLSTM is formed by



**Fig. 3** Architectural design of the single-layer BiLSTM-attention network for dynamic phase correction and salient point weighting

concatenating the forward hidden state  $\vec{h}_t$  and backward hidden state  $\overleftarrow{h}_t$ ,  $\mathbf{h}_t = [\vec{h}_t, \overleftarrow{h}_t]$ .

To enhance the ability of the model to capture key lag features, an attention mechanism was introduced to assign adaptive weights to the BiLSTM hidden states. The attention weights  $\alpha_t$  and the context vector  $\mathbf{c}_t$  are calculated as:

$$\mathbf{u}_t = \tanh(\mathbf{W}_w \mathbf{h}_t + \mathbf{b}_w), \quad (7)$$

$$\alpha_t = \frac{\exp(\mathbf{u}_t^\top \mathbf{u}_c)}{\sum_t \exp(\mathbf{u}_t^\top \mathbf{u}_c)}, \quad (8)$$

$$\mathbf{c} = \sum_t \alpha_t \mathbf{h}_t, \quad (9)$$

where  $\mathbf{h}_t$  is the output hidden state from the BiLSTM layer at time  $t$ .  $\mathbf{W}_w$  and  $\mathbf{b}_w$  represent the trainable weight matrix and bias vector, respectively.  $\mathbf{u}_t$  is the hidden representation of  $\mathbf{h}_t$ , and  $\mathbf{u}_c$  is a randomly initialized context vector learned during training to measure the importance of each time step. The final context vector  $\mathbf{c}$  is computed as the weighted sum of all hidden states based on the attention scores  $\alpha_t$ .

An optimized single-layer BiLSTM-attention architecture was constructed based on the preprocessed dynamic phase data. To simulate continuous real-time monitoring, the input is defined as a sliding window tensor of historical  $\text{CO}_2$  observations (Fig. 3). The proposed architecture processes the input sliding window through three main components: 1) a single BiLSTM layer for bidirectional temporal feature extraction, 2) an attention layer that dynamically assigns higher connection weights to time steps with significant concentration changes, and 3) a fully connected module that maps the extracted features to the target concentration. To mitigate overfitting under limited data conditions, L2 regularization (coefficient=0.005) and dropout (rate=0.5) were applied.

### 3.4 Experimental setup and evaluation strategy

Model evaluation followed a hold-out validation scheme with offline training and online inference. During training, the 11 sequences were segmented into sliding windows to optimize the network weights. The model weights were fixed during testing. The model is then applied to independent test Sets (A and B) to simulate real-time inference in continuous data streams.

To quantitatively evaluate the phase-correction performance and noise-resistance capabilities of the models across different dynamic characteristics, three standard metrics were adopted: the mean absolute error (MAE), root mean square error (RMSE), and coefficient of determination ( $R^2$ ). Among them,  $R^2$  serves as the core metric for evaluating the overall goodness-of-fit, and is mathematically defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2}, \quad (10)$$

where  $y_i$  represents the actual ground truth concentration,  $\hat{y}_i$  is the predicted value, and  $\bar{y}_i$  is the mean of the actual ground-truth data.

Regarding the evaluation criterion, the theoretical maximum of  $R^2$  was 1.0, indicating perfect fit. Therefore, a value closer to 1.0 always represents better predictive performance.

Within this evaluation system, a comprehensive, multitiered experimental scheme was designed to rigorously validate the superiority and structural necessity of the proposed framework.

- 1) Baseline comparison (validation of deep learning vs. traditional methods): The proposed BiLSTM-attention framework, as a benchmark, was compared with a classical physics-based LTI inverse filtering method and a traditional machine learning support vector regression (SVR) model, demonstrating the advantages of deep learning in balancing time-lag correction and fluid noise suppression.
- 2) Ablation study (validating the necessity of attention): A controlled ablation experiment was performed to compare the proposed full framework with the BiLSTM-only model. This study isolates and quantifies the critical contribution of the attention layer in mitigating the sluggish response typical of recurrent networks when handling abrupt step changes in  $p\text{CO}_2$  concentration.
- 3) Downstream Kalman filter (KF) post-processing gain (validation of micro-jitter suppression): To fulfill the stringent data quality requirements of accurate marine carbon flux calculations, the necessity of a cascaded KF was evaluated. We quantitatively compared the raw output of the network with the KF-smoothed output to verify its ability to eliminate inherent high-frequency prediction micro-jitter without compromising the dynamically corrected phase.

## 4 Experimental results and analysis

### 4.1 Baseline comparison: deep learning vs. traditional methods

To evaluate the phase correction performance and noise robustness of the BiLSTM-attention framework, we conducted comparative experiments on two independent test sets: test Set A (falling trend) and test Set B (rising trend). The proposed framework is benchmarked against three representative baselines.

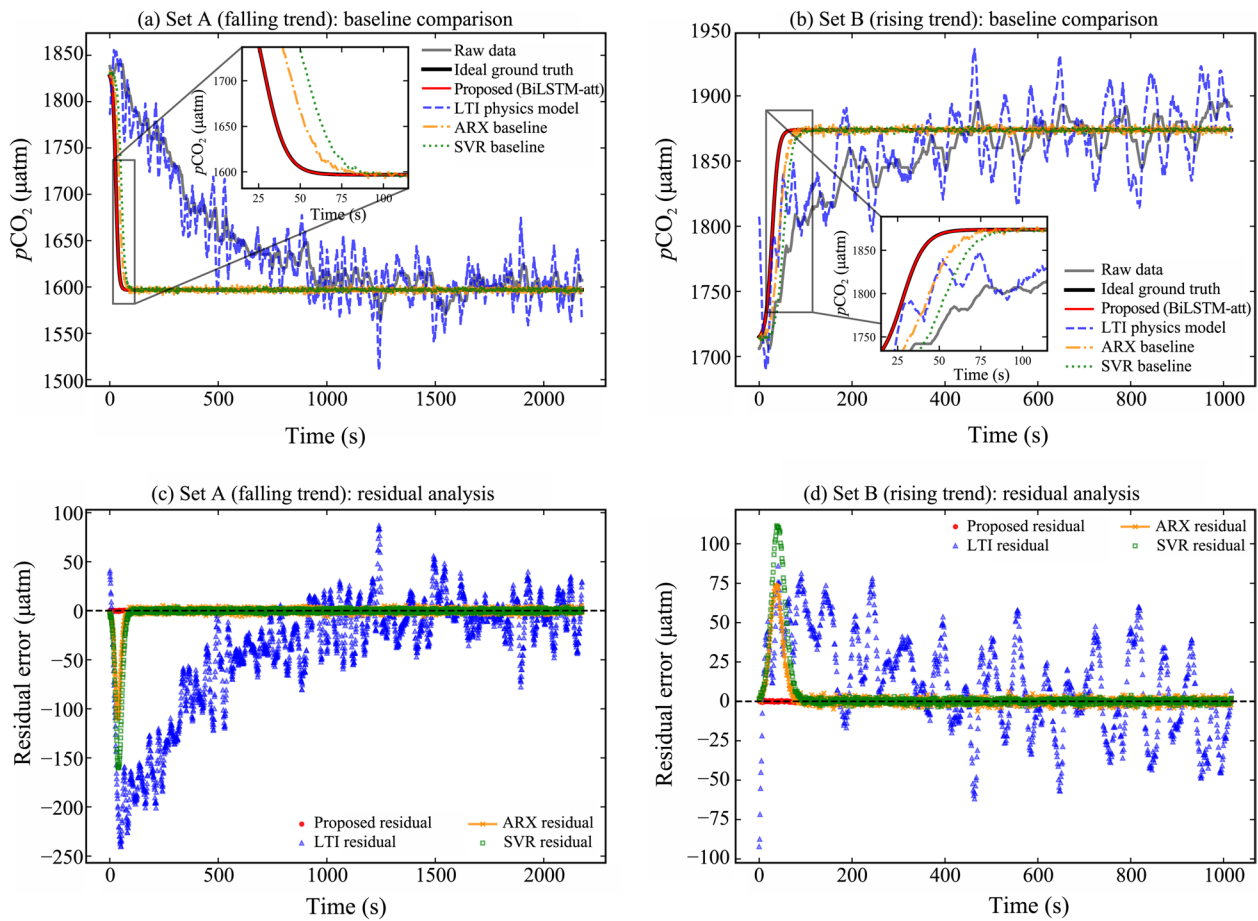
- 1) Linear time-invariant (LTI) model: a classical physics-based method for gradient inversion.
- 2) Auto regressive with exogenous input (ARX) model: which is a traditional linear autoregressive filtering approach.
- 3) Support vector regression (SVR): a conventional machine learning model lacking temporal memory.

Figure 4a and b show the prediction results for test Sets A and B, respectively, comparing the proposed framework with the LTI, ARX, and SVR models. Figure 4c and d present the corresponding residual error distributions. The ideal ground truth represents the theoretical concentration step change induced by the fluid-switching process described in Section 3.1.

As summarized in Table 1 and Fig. 4, the traditional baselines have notable limitations. Although the LTI model reduces the phase delay, its gradient-based formulation amplifies the high-frequency fluid noise, resulting in oscillatory residuals. The ARX and SVR models attenuate this noise, but fail to capture the nonlinear membrane permeation process, leading to a systematic bias and a rightward phase shift during transient dynamics. In contrast, the BiLSTM-attention framework closely tracks the ground truth, achieving  $R^2 > 0.99$  and maintaining an RMSE below  $2.0 \mu\text{atm}$  across both test sets. By adaptively weighting the salient features, the model effectively aligns the phase while preserving signal smoothness, thereby balancing lag correction and noise suppression.

### 4.2 Ablation study: crucial role of the attention mechanism

An ablation study was conducted to assess the contribution of the attention mechanism by comparing the proposed framework with a basic BiLSTM model. As illustrated by the transient response curves (Fig. 5), the model without the attention layer exhibited a pronounced delay during sharp concentration transitions, indicating that without adaptive weighting of the gradient changes, the base recurrent network suffers from a phase lag. Quantitative results further substantiate this observation. In test Set A (falling trend), removing the attention mechanism increases the RMSE from  $1.87 \mu\text{atm}$



**Fig. 4** Correction performance across multiple models

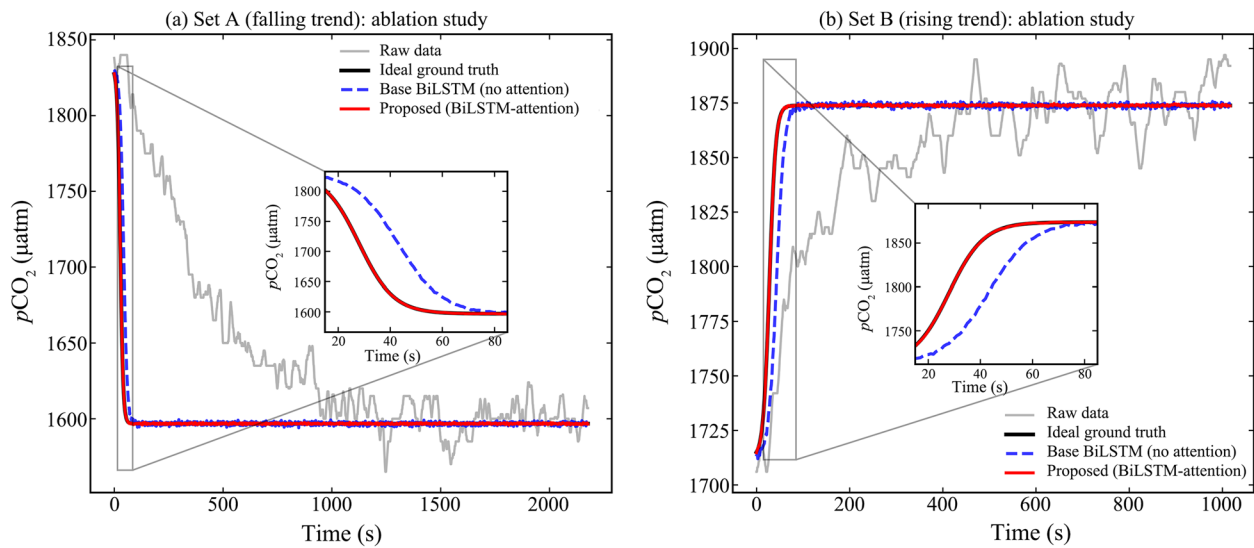
**Table 1** Summary of quantitative performance of different baseline comparisons

| Test set        | Method                      | RMSE (µatm) | MAE (µatm) | $R^2$ |
|-----------------|-----------------------------|-------------|------------|-------|
| Set A (falling) | Proposed (BiLSTM-attention) | 1.87        | 1.85       | 0.992 |
|                 | ARX baseline                | 15.20       | 12.30      | 0.880 |
|                 | LTI physics model           | 18.40       | 14.50      | 0.850 |
|                 | SVR baseline                | 22.30       | 18.50      | 0.710 |
| Set B (rising)  | Proposed (BiLSTM-attention) | 1.66        | 1.64       | 0.996 |
|                 | ARX baseline                | 17.10       | 14.20      | 0.860 |
|                 | LTI physics model           | 19.50       | 15.80      | 0.830 |
|                 | SVR baseline                | 24.50       | 20.10      | 0.680 |

to 11.50 µatm, and reduces the  $R^2$  from 0.992 to 0.920. A similar trend is observed in test Set B (rising trend), where the ablated model yields an RMSE of 12.30 µatm and an  $R^2$  of 0.910, compared with 1.66 µatm and 0.996 for the full framework. These findings demonstrate that the attention mechanism is essential for mitigating the

sensor hysteresis and accurately capturing rapid dynamic variations.

The performance comparison in Fig. 5a and b further shows that removing the attention layer (ablation: BiLSTM only) leads to a phase misalignment and increased lag during sharp  $p\text{CO}_2$  concentration step changes.



**Fig. 5** Ablation study on the structural necessity of the attention mechanism

#### 4.3 Downstream micro-jitter suppression via KF

While the BiLSTM-attention model effectively corrects macroscopic phase distortions, it introduces minor high-frequency stochastic micro-jitters in the prediction signals. To satisfy the data quality requirements for marine carbon flux estimation, a (KF) was employed as a downstream post-processing step.

This is shown in Fig. 6a and b for test Sets A and B, respectively. The zoomed-in insets highlight that the cascaded KF enhances signal stability while preserving the corrected transient phase.

The KF stage substantially suppressed the high-frequency micro-jitters, achieving a noise reduction of over 50% without degrading the phase alignment (Table 2, Fig. 6). In addition, this post-processing step significantly improved the dynamic response of the system, as quantified by  $t_{63}$  (the time required to reach 63.2% of the steady-state value). The response delay is reduced from 147.3 s to 18.2 s in the rising scenario (an 87.6% reduction), and from 273.3 s to 24.5 s in the falling scenario (a 91.0% reduction). Overall, this hybrid architecture, which combines deep learning-based phase correction with Kalman filtering, effectively resolves the trade-off between hysteresis compensation and noise amplification, enabling highly accurate continuous  $p\text{CO}_2$  measurements.

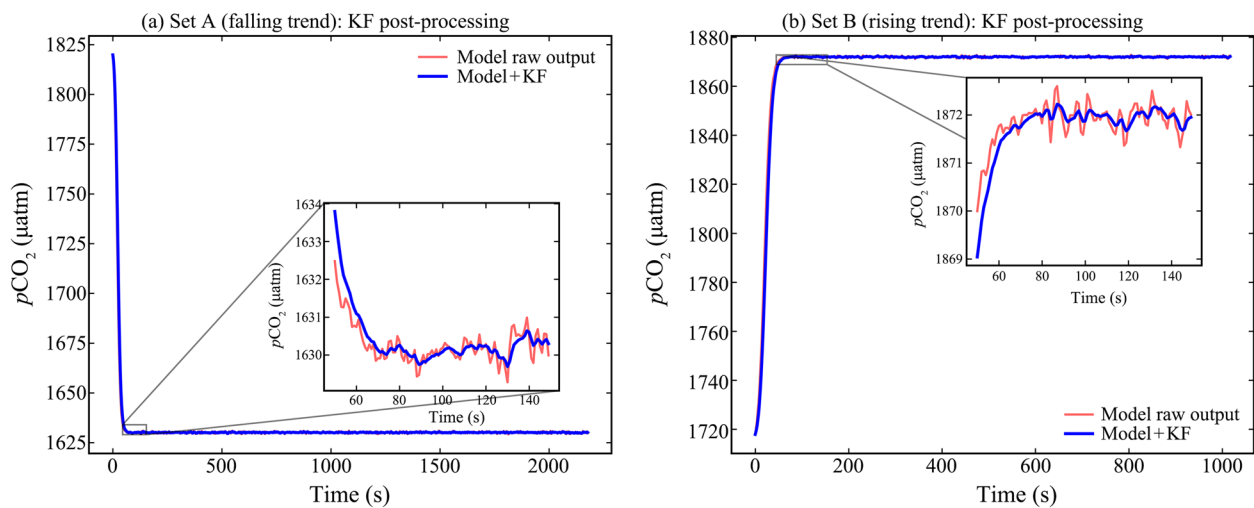
#### 4.4 Discussion

Acquiring large, high-quality, in situ marine datasets is both experimentally demanding and costly. This study demonstrates that a data-efficient modeling strategy can achieve accurate dynamic corrections. By extracting the transient phase features and incorporating regularization (L2 and dropout) within a lightweight single-layer

architecture, the model mitigates overfitting and achieves robust generalization ( $R^2$  and RMSE) on a limited dataset. This suggests that the proposed neural network is well-suited for resource-constrained marine sensing applications and requires substantially less training data than highly parameterized architectures such as transformers.

While Section 4 highlighted the limitations of the physical LTI and linear ARX models, the present ablation study clarifies the internal contribution of the proposed architecture. Conventional machine learning approaches such as SVR lack the temporal memory required to capture complex fluid dynamics. Although standard BiLSTM can model global temporal trends, it still exhibits a noticeable delay, often exceeding 40 s in our experiments during abrupt  $p\text{CO}_2$  concentration changes. The attention mechanism alleviates this limitation by selectively weighting transient features, thereby enabling the network to focus on rapid gradient variations. This targeted extraction effectively reduced the residual hysteresis, which is typical of standard recurrent networks.

Despite these promising results, the relatively small dataset (13 sequences) introduced inherent limitations. The model may partially overfit laboratory-specific noise characteristics rather than fully capturing universal permeation dynamics. In addition, the sensor response kinetics are intrinsically coupled with environmental variables such as temperature, salinity, and flow rate. Although the current framework has been validated under controlled conditions, its robustness under highly variable or extreme environmental regimes remains to be established. Therefore, future work should focus on expanding the multiparameter



**Fig. 6** Effect of KF post-processing on the high-frequency stochastic micro-jitters introduced by the neural network

**Table 2** Quantitative gain of the cascaded KF post-processing

| Test set        | Metric                            | Raw network output | KF smoothed output | Improvement     |
|-----------------|-----------------------------------|--------------------|--------------------|-----------------|
| Set A (falling) | Signal noise level                | 1.1271             | 0.5317             | 52.8% reduction |
|                 | System response time ( $t_{63}$ ) | 273.3 s            | 24.5 s             | 91.0% reduction |
| Set B (rising)  | Signal noise level                | 1.2396             | 0.4309             | 65.2% reduction |
|                 | System response time ( $t_{63}$ ) | 147.3 s            | 18.2 s             | 87.6% reduction |

dataset and incorporating field observations to ensure reliable performance under realistic oceanographic conditions.

## 5 Conclusions

To address sensor response hysteresis and high-frequency fluid noise in continuous marine CO<sub>2</sub> monitoring, this study proposes a correction framework that integrates a BiLSTM-attention network with Kalman filtering. The main findings of this study are as follows.

First, the proposed framework mitigates the noise amplification typically observed in LTI-based inverse filtering. It accurately reconstructs phase dynamics on independent test sets with high  $R^2$  and low RMSE, and recovers the dynamic  $p\text{CO}_2$  signal with minimal phase distortion.

Second, the ablation results confirm that the attention mechanism is essential. Emphasizing transient features reduces the response delay observed in standard recurrent networks during abrupt concentration step changes.

Finally, coupling the neural network with a downstream Kalman filter effectively suppresses high-frequency prediction noise. This cascaded approach

reduces the signal variance by more than 50% and shortens the system response time ( $t_{63}$ ) by 91.0% and 87.6% for the rising and falling  $p\text{CO}_2$  scenarios, respectively.

This paper presents a data-efficient deep learning solution for marine sensor calibration. By integrating temporal modeling with adaptive feature weighting and statistical filtering, the proposed framework enhances temporal resolution and data fidelity for in situ marine biogeochemical observations.

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## Additional information

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## Authors' contributions

All authors contributed to the conception and design of this study. Meng Li and Mengtie Du performed material preparation, data collection, and analysis. The first draft of the manuscript was written by Mengtie Du. Xuezu Wang

contributed to manuscript polishing and grammar/wording corrections. Yan-zhen Gu coordinated the study and provided valuable suggestions. Peiliang Li provided valuable suggestions. All authors commented on the previous versions of the manuscript. All the authors have read and approved the final version of the manuscript.

#### Data availability

Data and references presented in this paper are available from the corresponding author upon request.

#### Declarations

#### Ethics approval and consent to participate

Not applicable.

#### Consent for publication

Informed consent for publication was obtained from all participants.

#### Competing interests

The authors declare that they have no competing interests.

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